Moinuddin Mullan

Full-Stack Developer | AI/ML Developer

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Open to Remote, Relocate
LinkedIn

Profile

Computer science student pursuing Master of Computer Applications (MCA) with a strong foundation in full-stack development, AI/ML. Skilled in Java, Python, JavaScript, React, Node.js, PyTorch, and TensorFlow, with practical exposure to ML pipelines. Interested in applying technical skills in real-world software development and AI/ML engineering roles.

Education

Gujarat Secondary and Higher Secondary Education Board (GSEB) 2019 – 2020
Secondary School Certificate (SSC) — Class X Percentage: 67.66%

Gujarat Secondary and Higher Secondary Education Board (GSEB) 2021 – 2022
Higher Secondary Certificate (HSC) — Class XII, Science Stream Percentage: 57.14%

Veer Narmad South Gujarat University (VNSGU), Valsad, India Jul 2022 – Apr 2025
Bachelor of Computer Applications (BCA) CGPA: 8.75 / 10

Shrimad Rajchandra Institute of Management & Computer Application, UTU, Bardoli Jul 2025 – Present
Master of Computer Applications (MCA) Sem 1 SGPA: 9.64 / 10

Skills

Programming Languages: Java, Python, JavaScript, C, C++
Frontend: React.js, HTML, CSS, JavaScript
Mobile Development: Flutter, Dart
Backend: Node.js, Express.js, REST APIs, FastAPI
Databases: MySQL, MongoDB, SQL
Version Control: Git / GitHub
AI / ML Frameworks: PyTorch, TensorFlow, Scikit-learn, HuggingFace, LangChain
Core CS: Data Structures & Algorithms (DSA)
Languages: English, Gujarati, Hindi

Courses & Certifications

Getting Started with AI on Jetson Nano NVIDIA | Aug 2025
Credential ID: DiDHCiWZTFymu35LfnqJpA View Certificate

Participant — 7th International Conference on Soft Computing & its Engineering Applications (icSoftComp2025)
CHARUSAT | Springer | Dec 2025
Held at Hanoi, Vietnam, Dec 09–11, 2025 View Certificate

Building RAG Agents with LLMs NVIDIA | Nov 2026
Credential ID: mUCC5YjBQD6FZQq4EAC-sg View Certificate

Digital Badges



Microsoft Pro Badge: GitHub Copilot — Proficient
Microsoft | Nov 2026 View Badge ↗
Credential ID: 8dfe3594-fb30-4d0a-afd7-592d342878a0



Building RAG Apps Using MongoDB
MongoDB | Nov 2026 View Badge ↗
Credential ID: c2fe107e-3fe0-44c7-b0cd-8357a78c517e

Research / Review Papers

Multimodal and Deep Learning-Based Thermal Image Super-Resolution: A Review

Collaborated within a **six-member** research team with Professor. A review tracing thermal IR super-resolution from classical interpolation to CNN, GAN, Transformer, and Mamba-based architectures; covers optical-guidance strategies, multimodal fusion, degradation modelling, and benchmarks across medical, defense, remote sensing, and autonomous-driving domains.

Under Review / Manuscript in Preparation *View Manuscript* ↗

A Voice-Based AI Helpdesk and Appointment Management Simulator for Automotive Service Centres

A fully offline, open-source voice helpdesk simulator chaining Silero VAD, Faster-Whisper ASR, a rule-based fuzzy-matching intent classifier, a finite-state booking dialogue manager, and Facebook MMS TTS; achieves 78.3% weighted intent accuracy (60 typed utterances) and 3.9 s average latency on CPU-only hardware.

Under Review / Manuscript in Preparation *View Manuscript* ↗

Projects

MCA Projects

iOnYou — Smart Attendance Monitoring: A Cost-Optimized IoT Approach 2026

Tech Stack: React.js, Node.js, Express.js, MongoDB, ESP32-CAM, MFRC522 RFID Live Demo ↗ GitHub ↗

- Built a **dual-verification attendance system** combining ESP32-CAM face recognition and MFRC522 RFID scanning; attendance is recorded only when both biometric and physical credentials match the same student profile, eliminating proxy attendance.
- Developed a **MERN stack web portal** with role-based access for admin and faculty, supporting real-time attendance monitoring, student profile CRUD, and downloadable CSV/PDF attendance reports.

Hackathon Projects

Optical-Guided Super-Resolution for Thermal IR Imagery — SIH 2025 2025

Tech Stack: Python, PyTorch, Rasterio, Scikit-image **Theme:** Space Technology | ISRO Kaggle Model ↗

- Selected problem statement **SIH25171** (team *Thermal Titans*): enhancing low-resolution thermal IR imagery using high-resolution optical images as a spatial guidance signal, under the Space Technology / ISRO theme.
- Implemented a **10-block residual CNN** (VDSR/EDSR-inspired) trained on Landsat OLI+TIRS data using a patch-based `rasterio` pipeline with 4× bicubic degradation to simulate real sensor resolution gaps.
- Replaced MSE with a combined **L1 + Sobel gradient loss** to reduce output blur; added patience-based early stopping and best-checkpoint saving on the SSL4EO-L benchmark.

Freelancing Projects

PR Monitoring — Real-Time Server Monitoring Dashboard 2026

Tech Stack: React.js, Node.js, Express.js, MongoDB Atlas, Python (psutil), JWT, bcrypt, Recharts, Google Chat Webhooks

Live Demo ↗

- Built a **Live server monitoring platform** (React, Node.js/Express, MongoDB Atlas) deployed on **AWS EC2** , providing real-time visibility into CPU, memory, disk, I/O, and file-descriptor usage across multiple production servers.
- Developed a **Python agent** (psutil) that auto-detects hardware specs and streams live metrics every 10–30 s via JWT-authenticated REST; engineered **Google Chat webhook alerting** with 10-minute per-metric cooldown to prevent notification floods.
- Built **Zabbix-style Recharts dashboards** with drag-to-zoom, custom date-range filtering, and live 10-second polling; extended the platform to surface **vehicle-fleet activity** (total/inactive counts) from an external MySQL source.

Personal Projects

CampusNav — Campus Navigation Application 2026
Tech Stack: Flutter, Dart GitHub ↗

- Designed and developed a cross-platform campus navigation app for students to explore and navigate within their campus premises.
- Built both a mobile application and a companion web page using Flutter and Dart, providing a unified experience across devices.
- Implemented interactive campus maps with location-based guidance to help students locate classrooms, facilities, and key landmarks.

AutoCare AI — AI-Powered Vehicle Service Booking Bot 2026
Tech Stack: React.js, Node.js, Express.js, MongoDB Atlas, Python GitHub ↗

- Built a full-stack web application featuring an AI bot that communicates with users through both **chat and voice** to understand their vehicle service requirements.
- Developed the frontend using React.js and the backend API with Node.js and Express.js, with MongoDB Atlas as the cloud database for storing bookings and user data.
- Implemented the AI conversational model in Python, enabling the bot to process user input and automatically create or store vehicle service bookings.

DocuMind — AI Document Q&A Assistant 2026
Tech Stack: Flutter, Dart, MongoDB Atlas, Python GitHub ↗

- Developed a Flutter-based mobile application that allows users to upload multiple PDFs, Word documents, and research papers for AI-powered question answering.
- Integrated a Python AI model that processes uploaded files and provides accurate, context-aware answers sourced directly from the user's documents.
- Used MongoDB Atlas as the database to store uploaded document metadata and user query history, with a clean Flutter/Dart frontend for seamless interaction.

YouTube Comment Sentiment Analyzer Sep 2026
Tech Stack: React.js, JavaScript, Python, FastAPI, MongoDB Atlas, scikit-learn, NLP GitHub ↗ Live Demo ↗

- Built a **React.js web application** that classifies YouTube comments into **Positive, Neutral, and Negative** sentiments using a machine learning model trained on a 45K-comment Kaggle dataset.
- Developed **FastAPI REST APIs** for single and batch comment analysis using **TF-IDF** and **Logistic Regression**.
- Integrated **MongoDB Atlas** to store analysis history and created interactive dashboards to visualize sentiment distribution and prediction results; deployed the frontend on **Netlify** and the backend on **Render**.

Telegram Bot Projects

VisionLab Bot — AI Vision MultiTool Telegram Bot 2026
Tech Stack: Python, python-telegram-bot, BLIP (HuggingFace), YOLOv8, DeepFace, EasyOCR, OpenCV, rembg GitHub ↗

- Built an **8-feature AI image-processing Telegram bot** supporting image captioning (BLIP), OCR (EasyOCR), background removal (U^2-Net/rembg), face detection (OpenCV Haar), object detection (YOLOv8), emotion analysis (DeepFace), cartoon effects, and image enhancement.
- Implemented a **modular handler architecture** (one file per feature) with inline keyboard menus, per-user mode state, and an image router that dispatches uploads to the correct AI pipeline automatically.
- Added a **rule-based conversational layer** with fuzzy-matched triggers across 13 intent categories, giving the bot natural small-talk responses alongside its core image-processing capabilities.